

Interior Gateway Routing Protocol (IGRP)

Interior Gateway Routing Protocol (IGRP) is a distance vector interior gateway protocol (IGP) developed by Cisco. It is used by routers to exchange routing data within an autonomous system.

IGRP is a proprietary protocol. IGRP was created in part to overcome the limitations of RIP (maximum hop count of only 15, and a single routing metric) when used within large networks. IGRP supports multiple metrics for each route, including bandwidth, delay, load, and reliability; to compare two routes these metrics are combined together into a single metric, using a formula which can be adjusted through the use of pre-set constants. By default, the IGRP composite metric is a sum of the segment delays and the lowest segment bandwidth. The maximum configurable hop count of IGRP-routed packets is 255 (default 100), and routing updates are broadcast every 90 seconds (by default). IGRP uses protocol number 9 for communication.

IGRP is considered a classful routing protocol. Because the protocol has no field for a subnet mask, the router assumes that all subnetwork addresses within the same Class A, Class B, or Class C network have the same subnet mask as the subnet mask configured for the interfaces in question. This contrasts with classless routing protocols that can use variable length subnet masks. Classful protocols have become less popular as they are wasteful of IP address space.

Enhanced Interior Gateway Routing Protocol (EIGRP)

Enhanced Interior Gateway Routing Protocol (EIGRP) is an advanced distance-vector routing protocol that is used on a computer network for automating routing decisions and configuration. The protocol was designed by Cisco Systems as a proprietary protocol, available only on Cisco routers. Partial functionality of EIGRP was converted to an open standard in 2013.

EIGRP is used on a router to share routes with other routers within the same autonomous system. Unlike other well-known routing protocols, such as RIP, EIGRP only sends incremental updates, reducing the workload on the router and the amount of data that needs to be transmitted.

EIGRP replaced the Interior Gateway Routing Protocol (IGRP) in 1993. One of the major reasons for this was the change to classless IPv4 addresses in the Internet Protocol, which IGRP could not support.

Overview

Almost all routers contain a routing table that contains rules by which traffic is forwarded in a network. If the router does not contain a valid path to the destination, the traffic is discarded. EIGRP is a dynamic routing protocol by which routers automatically share route information. This eases the workload on a network administrator who does not have to configure changes to the routing table manually.

In addition to the routing table, EIGRP uses the following tables to store information:

- **Neighbor Table:** The neighbor table keeps a record of the IP addresses of routers that have a direct physical connection with this router. Routers that are connected to this router indirectly, through another router, are not recorded in this table as they are not considered neighbors.
- **Topology Table:** The topology table stores routes that it has learned from neighbor routing tables. Unlike a routing table, the topology table does not store all routes, but only routes that have been

determined by EIGRP. The topology table also records the metrics for each of the listed EIGRP routes, the feasible successor and the successors. Routes in the topology table are marked as "passive" or "active". Passive indicates that EIGRP has determined the path for the specific route and has finished processing. Active indicates that EIGRP is still trying to calculate the best path for the specific route. Routes in the topology table are not usable by the router until they are inserted into the routing table. The topology table is never used by the router to forward traffic. Routes in the topology table will not be inserted into the routing table if they are active, are a feasible successor, or have a higher administrative distance than an equivalent path.

Information in the topology table may be inserted into the router's routing table and can then be used to forward traffic. If the network changes (for example, a physical link fails or is disconnected), the path will become unavailable. EIGRP is designed to detect these changes and will attempt to find a new path to the destination. The old path that is no longer available is removed from the routing table. Unlike most distance vector routing protocols, EIGRP does not transmit all the data in the router's routing table when a change is made, but will only transmit the changes that have been made since the routing table was last updated. EIGRP does not send its routing table periodically, but will only send routing table data when an actual change has occurred. This behavior is more inline with link-state routing protocols, thus EIGRP is mostly considered a hybrid protocol.

When a router running EIGRP is connected to another router also running EIGRP, information is exchanged between the two routers. They form a relationship, known as an adjacency. The entire routing table is exchanged between both routers at this time. After the exchange has completed, only differential changes are sent.

EIGRP is often considered a hybrid protocol because it also sends link state updates when link states change.

Features

EIGRP supports the following features:

- Support for Classless Inter-Domain Routing (CIDR) and variable length subnet masking. Routes are not summarized at the classful network boundary unless auto summary is enabled.
- Support for load balancing on parallel links between sites.
- The ability to use different authentication passwords at different times.
- MD5 and SHA-2 authentication between two routers.
- Sends topology changes, rather than sending the entire routing table when a route is changed.
- Periodically checks if a route is available, and propagates routing changes to neighboring routers if any changes have occurred.
- Runs separate routing processes for Internet Protocol (IP), IPv6, IPX and AppleTalk, through the use of protocol-dependent modules (PDMs).
- Backwards compatibility with the IGRP routing protocols.